

Introduction

This project develops an EOG-based hands-free computer control system that detects eye blinks and movements to perform laptop actions. Using AD8232, ESP32, and Python, the system aims to improve accessibility for users with limited motor abilities.

Objective

- ❑ Develop an EOG-based hands-free laptop control system.
- ❑ Detect eye blinks and movements using **AD8232** and **ESP32**.
- ❑ Translate eye signals into computer commands.
- ❑ Improve accessibility through low-cost assistive technology.

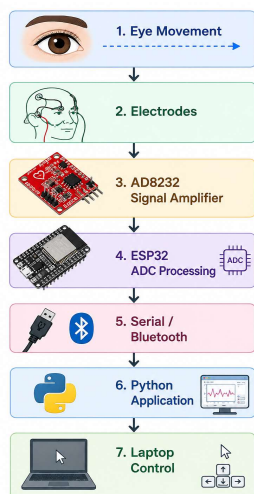
Motivation

- ❑ Traditional computer systems require hand-based input, which can be difficult for people with motor disabilities.
- ❑ Eye movements and blinks can provide an alternative natural method of interaction.
- ❑ A low-cost EOG-based system can improve accessibility and independence for users.
- ❑ This project explores affordable bio-signal technology for practical Human-Computer Interaction (HCI).

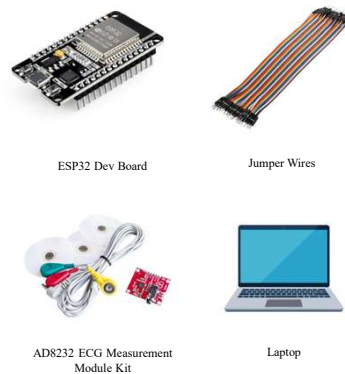


Figure 1: Assistive technology enabling accessible computer control

System Architecture



Hardware Setup

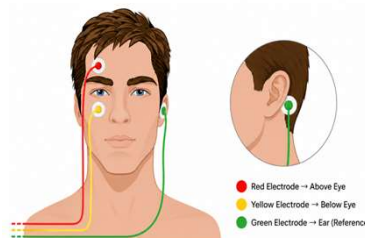


Pin Configuration

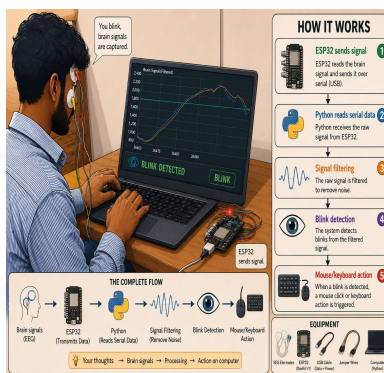
Table 1: ECG Sensor Module Wiring Configuration

AD8232 Pin	Connection	ESP32 Pin
3.3V	Red	3.3V
GND	Black	GND
OUTPUT	Green	GPIO34
LO+	Blue	GPIO32
LO-	Yellow	GPIO33

Electrode Placement



Software Workflow



Blink Mapping

Table 2: Blink Pattern to Laptop Command Mapping

Pattern	Action
Single Blink	Left Click
Double Blink	Right Click
Triple Blink	Shortcut
Look Left	Scroll Left
Look Right	Scroll Right

Signal Visualization

- ❑ Real-time EOG signals are processed and classified by the ESP32.
- ❑ Detected eye movements and blink patterns are displayed through the Arduino Serial Monitor.



Figure 2: Serial monitor output showing detected eye commands

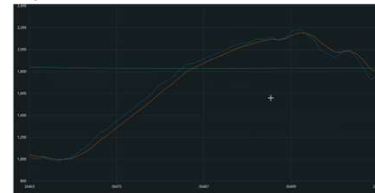


Figure 3: Real-Time EOG Signal Waveform

Result

- ❑ Successful blink detection
- ❑ Real-time response
- ❑ Hands-free laptop control achieved
- ❑ Low-cost implementation

Challenges & Future Work

The current system faces challenges including noise interference, electrode placement sensitivity, and the need for user-specific calibration. Future improvements may include wireless BLE communication, improved signal filtering, machine learning-based blink classification, and mobile device control integration.

Conclusion

This project demonstrates that low-cost EOG signals can enable practical hands-free computer interaction for accessibility applications.

References

- [1] C. Belkhiria, A. Boudir, C. Hurter, and V. Peysakhovich, "EOG-Based Human-Computer Interface: 2000–2020 Review," *Sensors*, vol. 22, no. 13, p. 4914, 2022, doi: 10.3390/s22134914.
- [2] L. Tao, Y. Liu, W. Chen, and C. Chen, "Review of electrooculography-based human-computer interaction: Recent technologies, challenges and future trends," *Connected Health and Telemedicine*, 2023.